

# Formula Sheet

2 × 2 Matrix inverse:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Quadratic equation  $ax^2 + bx + c = 0$  is solved by:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solution for n-dimensional LTI systems

$$x(t) = e^{A(t-t_0)} x_0 + \int_{t_0}^t e^{A(t-\tau)} B u(\tau) d\tau$$

Euler method:

$$x(t_{k+1}) \approx x(t_k) + \underbrace{(t_{k+1} - t_k)}_{=:h} f(x(t_k))$$

Heun Method:

$$\begin{aligned} x(t_{k+1}) &\approx x(t_k) + h \frac{1}{2} (K_1 + K_2) \\ K_1 &= f(x(t_k)) \\ K_2 &= f(x(t_k) + hK_1) \end{aligned}$$

Runge-Kutta Method:

$$\begin{aligned} x(t_{k+1}) &\approx x(t_k) + h \frac{1}{6} (K_1 + 2K_2 + 2K_3 + K_4) \\ K_1 &= f(x(t_k)) \\ K_2 &= f(x(t_k) + \frac{1}{2}hK_1) \\ K_3 &= f(x(t_k) + \frac{1}{2}hK_2) \\ K_4 &= f(x(t_k) + hK_3) \end{aligned}$$

Eigenvalues and eigenvectors of a matrix

$$A v = \lambda v,$$

Unity feedback connection of two systems:

$$\frac{G_1 G_2}{1 \pm G_1 G_2}$$

Transfer function of open-loop system:

$$G_c G_p,$$

Transfer function of closed-loop system:

$$\frac{G_c G_p}{1 + G_c G_p},$$

where  $G_c$  is the controller transfer function and  $G_p$  is the plant transfer function.

Conversion to transfer function:

$$G(s) = C(sI - A)^{-1} B + D$$

Controllability matrix:

$$Q_r = [B \quad AB \quad A^2 B \quad \dots \quad A^{n-1} B]$$

Observability matrix:

$$Q_o = \begin{bmatrix} C^T & A^T C^T & (A^T)^2 C^T & \dots & (A^T)^{n-1} C^T \end{bmatrix}$$

The Taylor series of the nonlinear system  $\dot{x}(t) = f(x(t))$  for the linearization points  $x^*$  is

$$\dot{x}(t) = f(x^*) + \underbrace{\frac{\partial}{\partial x} f(x, u) \Big|_{x=x^*, u=u^*}}_{=:A} (x(t) - x^*) + \dots$$

Initial and final value theorem:

$$\begin{aligned} \lim_{t \rightarrow 0} f(t) &= \lim_{s \rightarrow \infty} s F(s) \\ \lim_{t \rightarrow \infty} f(t) &= \lim_{s \rightarrow 0} s F(s) \end{aligned}$$

Laplace transform:

$$\begin{aligned}
1 &\circ \bullet \frac{1}{s} \\
e^{kt} &\circ \bullet \frac{1}{(s-k)} \\
t^n &\circ \bullet \frac{n!}{s^{n+1}} \text{ for } n = 0, 1, 2, \dots \\
t^n e^{-at} &\circ \bullet \frac{n!}{(s+a)^{n+1}} \text{ for } n = 0, 1, 2, \dots \\
(1-at)e^{-at} &\circ \bullet \frac{s}{(s+a)^2} \\
\sin(at) &\circ \bullet \frac{a}{s^2 + a^2} \\
\cos(at) &\circ \bullet \frac{s}{s^2 + a^2}
\end{aligned}$$

For  $G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$ , the state-space representation is:

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$C = [b_n - a_n b_0, b_{n-1} - a_{n-1} b_0, \dots, b_1 - a_1 b_0]$$

$$D = [b_0]$$

Matrix Riccati equation

$$-PA - A^T P + PBR^{-1}B^T P = Q$$

Incidence Matrix

$$\mathbf{N} = \mathbf{N}^+ - \mathbf{N}^-$$

with

$$n_{ij}^+ = \begin{cases} 1 & \text{if } (T_j, P_i) \in \mathcal{POST} \\ 0 & \text{otherwise} \end{cases}$$

$$n_{ij}^- = \begin{cases} 1 & \text{if } (P_i, T_j) \in \mathcal{PRE} \\ 0 & \text{otherwise} \end{cases}$$

DeMorgan

$$\neg(a \wedge b) = \neg a \vee \neg b$$

$$\neg(a \vee b) = \neg a \wedge \neg b$$

Temporal logics

$$true = p \vee \neg p$$

$$false = \neg true$$

$$\phi_1 \wedge \phi_2 = \neg(\neg\phi_1 \vee \neg\phi_2)$$

$$\phi_1 \implies \phi_2 = \neg\phi_1 \vee \phi_2$$

$$\phi_1 R \phi_2 = \neg(\neg\phi_1 U \neg\phi_2)$$

$$F(\phi) = true U \phi$$

$$G(\phi) = \neg F \neg \phi$$

$$AX(\phi) = \neg EX(\neg\phi)$$

$$EF(\phi) = E[true U \phi]$$

$$AG(\phi) = \neg EF(\neg\phi)$$

$$AF(\phi) = \neg EG(\neg\phi)$$

$$A[\phi_1 U \phi_2] = \neg E[\neg\phi_2 U (\neg\phi_1 \wedge \neg\phi_2)] \wedge \neg EG \neg\phi_2$$

$$A[\phi_1 R \phi_2] = \neg E[\neg\phi_1 U \neg\phi_2]$$

$$E[\phi_1 R \phi_2] = \neg A[\neg\phi_1 U \neg\phi_2]$$

Useful Theorems

$$a, a \implies b \vdash b \text{ (modus ponens)}$$

$$\neg a, a \vee b \vdash b$$

$$\neg b, a \implies b \vdash \neg a \text{ (modus tollens)}$$

$$a \vee b, a \implies c, b \implies c \vdash c$$

$$a \implies b, b \implies c \vdash a \implies c$$

$$a, a \implies b, b \implies c \vdash c$$

LMIs for checking stability of switched linear systems

$$P > 0$$

$$\forall i : (A^{(i)})^T P + P A^{(i)} < 0$$

Infeasibility Test

If there exist  $M$  positive definite matrices  $R^{(i)} > 0$  ( $M$ : number of locations) such that

$$\sum_{i=1}^M R^{(i)} (A^{(i)})^T + A^{(i)} R^{(i)} > 0$$

then there is no  $P > 0$  such that

$$\forall i \in \{1, \dots, M\} : (A^{(i)})^T P + P A^{(i)} < 0$$